`std::call_once` is used with an instance of `std::once_flag` to ensure that a particular function is called exactly once, even if multiple threads invoke the call concurrently.

Declaration

```
template<typename Callable,typename... Args>
void call_once(std::once_flag& flag,Callable func,Args args...);
```

Preconditions

The expression `INVOKE(func,args)` is valid for the supplied values of `func` and `args`. `Callable` and every member of `Args` are `MoveConstructible`.

Effects

Invocations of `std::call_once` on the same `std::once_flag` object are serialized. If there has been no prior effective `std::call_once` invocation on the same `std::once_flag` object, the argument `func` (or a copy thereof) is called as-if by *INVOKE(*`func,args`*)*, and the invocation of `std::call_once` is effective if and only if the invocation of `func` returns without throwing an exception. If an exception is thrown, the exception is propagated to the caller. If there has been a prior effective `std::call_once` on the same `std::once_flag` object, the invocation of `std::call_once` returns without invoking `func`.

Synchronization

The completion of an effective `std::call_once` invocation on a `std::once_flag` object happens-before all subsequent `std::call_once` invocations on the same `std::once_flag` object.

Throws